higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE

DIGITAL ELECTRONICS N4

(8080354)

13 April 2026 (X-paper)
09:00–12:00

Drawing instruments may be used. Calculators may NOT be used.

This question paper consists of 5 pages.

183Q1A2613

tvetpro.co.za



Try our Apps on Play Store

TVET Exam Papers NATED & NCV	**N1-N6** Mathematics NATED	**N1-N4** Engineering Science NATED	**N1-N6** Industrial Electronics NATED	**N4-N6** Electrotechnics NATED	**N4-N6** Mechanotechnics NATED
N5-N6 Power Machines NATED	**N4-N6** Quantity Surveying NATED	**N5-N6** Strength of Materials & Structures NATED	**N1-N3** Engineering Drawing NATED	**N3** Eletro-technology NATED	**N3** Mechano-technology NATED
N4-N6 Building Administration NATED	**N4-N6** Building& Structural Surveying NATED	**N4-N6** Building & Structural Construction NATED	**N1-N3** Building Science NATED	**N1-N3** Building Drawing NATED	**N4** Mechanical Draughting NATED
N1-N3 Electrical Trade Theory NATED	**N2-N3** Diesel Trade Theory NATED	**N5-N6** Strength of Materials & Structures NATED	**N3** Building & Civil Technology NATED	**N1-N2** Fitting and Machining Theory NATED	**N1-N4** Plating and Structural Steel Drawing NATED
N1-N3 Motor Trade Theory NATED	**N5-N6** Mechanical Drawing & Design NATED	**N1-N2** Plumbing Theory NATED	**N4-N6** Fault Finding & Protective Devices NATED	**N2-N3** Instrument Trade Theory NATED	**N5-N6** Engineering Physics NATED
N5-N6 Fluid Mechanics NATED	**N4-N6** Supervisory Management NATED	**N4-N6** Digital Electronics NATED	**N6** Control Systems NATED	**N1-N2** Bricklaying &Plastering Theory NATED	More…

DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
DIGITAL ELECTRONICS N4
TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer all the questions.
2. Read all the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Use only a black or blue pen.
5. Write neatly and legibly.

---

tvetpro.co.za

QUESTION 1

1.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.5) in the ANSWER BOOK.

1.1.1 Which code among the following code is weighted and non-self-complimentary?

A 8421 code
B 2421 code
C Excess-3 code
D Gray code

1.1.2 The hexadecimal equivalent of $123{,}4_8$ is …

A $53{,}8_{10}$.
B $83{,}578_{10}$.
C $53{,}8_{16}$.
D 83.578_{16}.

1.1.3 A simplified expression for $A + A\overline{B} + \overline{A}B$.

A 1
B A
C A + B
D $\overline{A} + B$

1.1.4 The output of … gate will be logic 1 if one or both inputs are logic 1.

A AND
B OR
C NOR
D NAND

1.1.5 Which one of the following is NOT a type of flip-flop?

A SR
B JK
C BD
D T

(5 × 1) (5)

tvetpro.co.za

1.2 Indicate whether the following statements are TRUE or FALSE by writing only 'True' or 'False' next to the question number (1.2.1–1.2.5) in the ANSWER BOOK.

1.2.1 A register is the group of flip-flops that are used for storing data in binary code.

1.2.2 The half adder is used to add three variables A, B and C.

1.2.3 If there are 3 flip-flops the counter counts up to 16.

1.2.4 An encoder converts decimal to binary.

1.2.5 SSI stands for Small Scale Integration.

(5 × 1) (5)
[10]

QUESTION 2

2.1 Complete the following calculations in the binary number system. All steps must be indicated clearly.

2.1.1 $-1011,101_2 - 1110,11_2$ (Make use of the one's complement method and convert the FINAL answer to the DECIMAL number.) (5)

2.1.2 $47,2_8 \div 44,25_{10}$ (Convert the FINAL answer to an OCTAL number.) (6)

2.1.3 $A, E_{16} \times 12, 1_8$ (Convert the FINAL answer to a HEXADECIMAL number.) (6)

2.2 Convert $101101,01_2$ to the EX3 code. (6)
[23]

tvetpro.co.za

QUESTION 3

3.1 Draw the logic circuit diagram of each of the following Boolean expressions. DO NOT change or simplify the expression.

3.1.1 $\left(\overline{B \oplus \overline{CD}}\right)\left(\overline{BC + \overline{BCD}}\right) = F$ (6)

3.1.2 $\overline{\left(AB \oplus \overline{BC}\right) \oplus \overline{ABC}} = F$ (5)

3.2 Simplify the expression below using Boolean algebra. Draw the simplified circuits using ONLY NAND gates.

3.2.1 $A(A + B + C)\left(\overline{B} + \overline{C}\right)\left(\overline{A} + B + C\right) = F$ (7)

3.2.2 $AB + \overline{\overline{A}B\overline{C} + A\overline{B} + \overline{A}C} = F$ (7)

[25]

QUESTION 4

4.1 Draw a neat logic circuit diagram for clocked D-latch flip-flop. (6)

4.2 Draw a neat, labelled sketch of a 3-bit asynchronous down counter. Use JK master slave F/F, also include a truth table and the timing diagram showing eight clock pulses. (12)

4.3 Differentiate between *fan in* and *fan out*. (4)

[22]

QUESTION 5

5.1 Draw a neat, labelled sketch of a 4:1 Multiplexer/Demultiplexer.

Clear show all input, output and control lines. (12)

5.2 Explain the operating principles of a ramp-type A/D converter. (8)

[20]

TOTAL: 100

ivetpro.co.za

Copyright reserved